

Practice Lab Test

• Model (JavaBean)

```
package com.lab.bean;
```

```
import java.io.Serializable;
```

```
public class UserBean implements Serializable {
```

```
// 1. letak attributes ikut kehendak soalan test
```

```
private String username;
```

```
private String fullname;
```

```
// 2. constructor kosong (wajib!)
```

```
public UserBean() {
```

```
}
```

```
// 3. Getter & Setter
```

```
public String getUsername() {
```

```
    return username;
```

```
}
```

```
public void setUsername(String username) {
```

```
    this.username = username;
```

```
}
```

```
public String getFullname() {
```

```
    return fullname;
```

```
}
```

```
public void setFullname(String fullname) {
```

```
    this.fullname = fullname;
```

```
}
```


• Database (DAO)

```
package com.lab.dao;
```

```
import com.lab.bean.UserBean;
```

```
import java.sql.*;
```

```
import java.util.ArrayList;
```

```
import java.util.List;
```

```
public class UserDao {
```

```
    // Fungsi pautan database
```

```
    private Connection getConnection() throws Exception {
```

```
        Class.forName("com.mysql.cj.jdbc.Driver");
```

pastikan guna
cj driver

```
        return DriverManager.getConnection("jdbc:mysql://localhost:3306/lab4_db", "root", "");
```

nama database

lewat password

```
    }
```

```
    // Contoh INSERT (CREATE)
```

nama model JavaBean

```
    public boolean insertUser(UserBean user) {
```

```
        String sql = "INSERT INTO students (matric_no, fullname) VALUES (?, ?)";
```

```
        try { Connection conn = getConnection(); PreparedStatement ps = conn.prepareStatement(sql);
```

```
            ps.setString(1, user.getUsername());
```

```
            ps.setString(2, user.getFullname());
```

```
            return ps.executeUpdate() > 0; // pulang true jika berjaya
```

```
        } catch (Exception e) {
```

```
            e.printStackTrace();
```

```
            return false;
```

```
        }
```

```
    }
```


// Contoh Retrieve list (READ)

```
public List<UserBean> getAllUsers() {  
  
    List<UserBean> list = new ArrayList<>();  
  
    String sql = "SELECT * FROM students";  
  
    try (Connection conn = getConnection();  
  
        PreparedStatement ps = conn.prepareStatement(sql);  
  
        ResultSet rs = ps.executeQuery()) {  
  
        while (rs.next()) {  
  
            UserBean u = new UserBean();  
  
            u.setUsername(rs.getString("matric_no"));  
  
            u.setFullname(rs.getString("fullname"));  
  
            list.add(u);  
  
        }  
  
    } catch (Exception e) {  
  
        e.printStackTrace();  
  
    }  
  
    return list;  
  
}
```


• Unified Controller (Servlet)

```
package com.lab.controller;
```

```
import com.lab.bean.UserBean;
```

```
import com.lab.dao.UserDAO;
```

```
import java.io.IOException;
```

```
import java.util.List;
```

```
import javax.servlet.ServletException;
```

```
import javax.servlet.annotation.WebServlet;
```

```
import javax.servlet.http.*;
```

```
@WebServlet("/UserServlet")
```

```
public class UserServlet extends HttpServlet {
```

```
    private UserDAO userDAO = new UserDAO();
```

```
    protected void processRequest (HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
```

```
        // tangkap parameter action
```

```
        String action = request.getParameter("action");
```

```
        try {
```

```
            if ("add".equals(action)) {
```

```
                // Selesaikan proses simpan data
```

```
                UserBean u = new UserBean();
```

```
                u.setUsername(request.getParameter("username"));
```

```
                u.setFullName(request.getParameter("full name"));
```

```
                userDAO.insertUser(u);
```

```
                response.sendRedirect("UserServlet?action=list");
```

```
            } else if ("list".equals(action)) {
```

```
                List<UserBean> users = userDAO.getAllUsers();
```

```
                request.setAttribute("userlist", users);
```

```
                request.getRequestDispatcher("viewUsers.jsp").forward(request, response);
```

```
            }
```

```
        } catch (Exception e) {
```

```
            throw new ServletException(e);
```

```
        }
```

```
    }
```

posisi sama dgn
name = "..."
dim tag <input>

lepas tambah, dra display

ambil data sesuai
dari database

simpan data request attribute
untuk JSP baca

forward ke halaman JSP,

@Override

```
protected void doGet (HttpServletRequest request, HttpServletResponse response)
```

```
    throws ServletException, IOException {
```

```
    processRequest (request, response);
```

```
}
```

@Override

```
protected void doPost (HttpServletRequest request, HttpServletResponse response)
```

```
    throws ServletException, IOException {
```

```
    processRequest (request, response);
```

```
}
```

```
}
```


• JSP

```
<%@page import = "com.lab.bean.UserBean"%>
```

```
<%@page import = "java.util.List"%>
```

```
<%@page contentType = "text/html" pageEncoding = UTF-8"%>
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head><title>User List </title></head>
```

```
<body>
```

```
<h2>List of Registered Users</h2>
```

```
<table border = "1">
```

```
<tr>
```

```
<th>Username</th>
```

```
<th>Fullname</th>
```

```
</tr>
```

```
<%
```

ambil data list yang servlet hunter

```
List<UserBean> list = (List<UserBean>) request.getAttribute("userList");
```

```
if (list != null) {
```

```
    for (UserBean u: list) {
```

```
%>
```

```
<tr>
```

```
<td><% = u.getUsername() %></td>
```

```
<td><% = u.getFullname() %></td>
```

```
</tr>
```

```
<%
```

```
}
```

```
}
```

```
%>
```

```
</table>
```

```
</body>
```

```
</html>
```